

Course 5333B

5 Credits: 4

Program Elective

Source-grounded

# Machine Vision

Study the official course objectives in the source syllabus.

## OFFICIAL SOURCE DECLARATION

### How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5333B is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 5333B.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

## COURSE DASHBOARD

### Official course information

#### Course information

| Item                 | Official information                     |
|----------------------|------------------------------------------|
| Programme            | Diploma in Automation and Robotics       |
| Course code          | 5333B                                    |
| Course title         | Machine Vision                           |
| Semester             | 5 Credits: 4                             |
| Course category      | Program Elective                         |
| Periods per week     | 4 (L:3 T:0 P:0) Periods per semester: 60 |
| Periods per semester | 60                                       |

| Item         | Official information                                 |
|--------------|------------------------------------------------------|
| Credits      | 4                                                    |
| Prerequisite | See the official syllabus PDF                        |
| Assessment   | Use the official assessment section reproduced below |

## Modules

4 official module sections detected.

## Topic entries

33 source-aligned topic entries retained.

## Study mode

Theory and application emphasis.

### STUDY METHOD

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

## Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

### LEARNING OUTCOMES

## Objectives, outcomes and mapping

### Course objectives

- Study the official course objectives in the source syllabus.

#### Course outcomes

| CO  | Official outcome                                                                 | Study level           |
|-----|----------------------------------------------------------------------------------|-----------------------|
| CO1 | 14 industry standards. To know about Image Representation, Matching and Applying | See official syllabus |
| CO2 | 15 Inference To expose the students to image feature detection and Applying      | See official syllabus |
| CO3 | 15 matching. Apply various transform tools like frequency domain Understanding   | See official syllabus |
| CO4 | 14 and affine transform.                                                         | See official syllabus |

#### CO-PO mapping evidence

| Row | Extracted official mapping |
|-----|----------------------------|
| CO1 | CO1                        |
| CO2 | CO2 2 2                    |
| CO3 | CO3 2 2                    |
| CO4 | CO4 2                      |

### COVERAGE AUDIT

## Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

#### Module coverage

| Module   | Official heading                                                                        | Topic entries | Hours        |
|----------|-----------------------------------------------------------------------------------------|---------------|--------------|
| Module 1 | Image Acquisition, Lenses and Cameras: Pinhole camera, Gaussian Optics, Depth Field,    | 7             | As specified |
| Module 2 | Image Fundamentals Images, regions, piecewise contours, Image Enhancement: contrast     | 9             | As specified |
| Module 3 | Image Transformations: Nearest-Neighbour Interpolation, Bilinear Interpolation, Bicubic | 4             | As specified |
| Module 4 | Image Segmentation: Region Growing, Edge Based approaches to segmentation, Graph-Cut,   | 13            | As specified |

#### MODULE 1

## Image Acquisition, Lenses and Cameras: Pinhole camera, Gaussian Optics, Depth Field,

This module is presented from the official syllabus scope for Machine Vision. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

### Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Image Acquisition, Lenses and Cameras: Pinhole camera, Gaussian Optics, Depth Field,

- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

## Image Acquisition, Lenses and Cameras: Pinhole camera, Gaussian Optics, Depth Field,

Image Acquisition, Lenses and Cameras: Pinhole camera, Gaussian Optics, Depth Field, is an official topic in Module 1 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Image Acquisition, Lenses and Cameras: Pinhole camera, Gaussian Optics, Depth Field, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, image acquisition, lenses and cameras: pinhole camera, gaussian optics, depth field, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Image Acquisition, Lenses and Cameras: Pinhole camera, Gaussian Optics, Depth Field, means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                        |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                              |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-ഇംഗ്ലീഷ് Image Acquisition, Lenses and Cameras: Pinhole camera, Gaussian Optics, Depth Field, ഫോക്കസ് ദൂരം, ഫോക്കസ് പോയിന്റ് definition/meaning ഇംഗ്ലീഷ്. ഫോക്കസ് പോയിന്റ് ഫോക്കസ് പോയിന്റ്, steps, application ഇംഗ്ലീഷ് ഫോക്കസ് പോയിന്റ് ഫോക്കസ് പോയിന്റ്. Technical terms English-ഇംഗ്ലീഷ് ഫോക്കസ് പോയിന്റ്.

## Telemetric lenses, Lens aberrations Cameras: CCD cameras, CMOS Camera, Colour cameras:

Telemetric lenses, Lens aberrations Cameras: CCD cameras, CMOS Camera, Colour cameras: is an official topic in Module 1 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Telemetric lenses, Lens aberrations Cameras: CCD cameras, CMOS Camera, Colour cameras: using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, telemetric lenses, lens aberrations cameras: ccd cameras, cmos camera, colour cameras: should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Telemetric lenses, Lens aberrations Cameras: CCD cameras, CMOS Camera, Colour cameras: means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                          |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.



**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-തെലിമെട്രിക് ലെൻസ്, ലെൻസ് അബറേഷൻസ് ക്യാമറാസ്: CCD ക്യാമറാസ്, CMOS ക്യാമറ, കോളർ ക്യാമറാസ്: നിർവ്വചനം/അർത്ഥം, ഘട്ടങ്ങൾ, പ്രയോഗങ്ങൾ. നിർവ്വചനം/അർത്ഥം, ഘട്ടങ്ങൾ, പ്രയോഗങ്ങൾ. Technical terms English-ൽ നിർവ്വചനം/അർത്ഥം.

## Single chip cameras, Three chip cameras, Camera performance parameters: noises, dynamic range,

Single chip cameras, Three chip cameras, Camera performance parameters: noises, dynamic range, is an official topic in Module 1 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Single chip cameras, Three chip cameras, Camera performance parameters: noises, dynamic range, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, single chip cameras, three chip cameras, camera performance parameters: noises, dynamic range, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Single chip cameras, Three chip cameras, Camera performance parameters: noises, dynamic range, means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                  |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                        |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-ഒരു Single chip cameras, Three chip cameras, Camera performance parameters: noises, dynamic range, ഓർഗനൈസേഷൻ ഓർഗനൈസേഷൻ definition/meaning ഓർഗനൈസേഷൻ. ഓർഗനൈസേഷൻ ഓർഗനൈസേഷൻ, steps, application ഓർഗനൈസേഷൻ ഓർഗനൈസേഷൻ ഓർഗനൈസേഷൻ. Technical terms English-ഒരു ഓർഗനൈസേഷൻ ഓർഗനൈസേഷൻ.

## Camera-Computer interfaces, Digital video signals: camera link, IEEE1394, USB2.0, USB 3

Camera-Computer interfaces, Digital video signals: camera link, IEEE1394, USB2.0, USB 3 is an official topic in Module 1 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Camera-Computer interfaces, Digital video signals: camera link, IEEE1394, USB2.0, USB 3 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, camera-computer interfaces, digital video signals: camera link, ieee1394, usb2.0, usb 3 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Camera-Computer interfaces, Digital video signals: camera link, IEEE1394, USB2.0, USB 3 means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                 |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-കാമറ-കമ്പ്യൂട്ടർ ഇന്റർഫേസ്, ഡിജിറ്റൽ വീഡിയോ സിഗ്നൽ: കാമറ ലിങ്ക്, IEEE1394, USB2.0, USB 3 സിംഗിൾ ലൈൻ ഡിഫിനിഷൻ/മീനിംഗ്. ഏതെങ്കിലും ഏതെങ്കിലും ഏതെങ്കിലും, steps, application ഏതെങ്കിലും ഏതെങ്കിലും ഏതെങ്കിലും. Technical terms English-ൽ എഴുതുക ഏതെങ്കിലും.

## vision

vision is an official topic in Module 1 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify vision using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, vision should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What vision means in the official course context                                  |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- vision പരസ്പരം വിരുദ്ധമായ വീക്ഷണങ്ങൾ definition/meaning പരസ്പരം വിരുദ്ധമായ വീക്ഷണങ്ങൾ. പരസ്പരം വിരുദ്ധമായ വീക്ഷണങ്ങൾ, steps, application പരസ്പരം വിരുദ്ധമായ വീക്ഷണങ്ങൾ. Technical terms English- പരസ്പരം വിരുദ്ധമായ വീക്ഷണങ്ങൾ.

## Explain different image smoothing algorithms used

Explain different image smoothing algorithms used is an official topic in Module 1 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain different image smoothing algorithms used using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain different image smoothing algorithms used should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                             |
|---------------------|---------------------------------------------------------------------------------------------|
| Definition/identity | What Explain different image smoothing algorithms used means in the official course context |
| Study lens          | Principle → components or stages → result → application                                     |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate           |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.



**Malayalam support:** Module 1- Explain different image smoothing algorithms used. definition/meaning. steps, application. Technical terms English-Malayalam.

## in computer vision

in computer vision is an official topic in Module 1 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify in computer vision using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, in computer vision should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What in computer vision means in the official course context                      |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- $\square$  in computer vision  $\square\square\square\square\square\square\square\square\square$   $\square\square\square\square$  definition/meaning  $\square\square\square\square\square\square\square\square\square\square$ .  $\square\square\square\square\square$   $\square\square\square\square\square$   $\square\square\square\square\square\square$ , steps, application  $\square\square\square\square\square$   $\square\square\square\square\square\square\square\square$   $\square\square\square\square\square$ . Technical terms English- $\square$   $\square\square\square\square$   $\square\square\square\square\square\square\square\square\square$ .

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Image Acquisition, Lenses and Cameras: Pinhole camera, Gaussian Optics, Depth Field,  
Telemetric lenses, Lens aberrations Cameras: CCD cameras, CMOS Camera, Colour cameras:  
Single chip cameras, Three chip cameras, Camera performance parameters: noises, dynamic range,  
Camera-Computer interfaces, Digital video signals: camera link, IEEE1394, USB2.0, USB 3  
vision

|                                                                 |                                                   |   |
|-----------------------------------------------------------------|---------------------------------------------------|---|
| C02: To know about Image Representation, Matching and Inference |                                                   |   |
| M2.01                                                           | Explain image enhancement techniques.             | 8 |
| Understanding                                                   |                                                   |   |
|                                                                 | Explain different image smoothing algorithms used |   |
| M2.02                                                           |                                                   | 7 |
| Understanding                                                   |                                                   |   |
|                                                                 | in computer vision                                |   |
|                                                                 | Series Test – I                                   | 1 |

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## MODULE 2

# Image Fundamentals Images, regions, piecewise contours, Image Enhancement: contrast

This module is presented from the official syllabus scope for Machine Vision. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Image Fundamentals Images, regions, piecewise contours, Image Enhancement: contrast
- Official duration: As specified
- Official topic entries captured: 9
- The full source excerpt is preserved below for coverage checking.

## Image Fundamentals Images, regions, piecewise contours, Image Enhancement: contrast

Image Fundamentals Images, regions, piecewise contours, Image Enhancement: contrast is an official topic in Module 2 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Image Fundamentals Images, regions, piecewise contours, Image Enhancement: contrast using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, image fundamentals images, regions, piecewise contours, image enhancement: contrast should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                               |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Image Fundamentals Images, regions, piecewise contours, Image Enhancement: contrast means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                       |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                             |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-ഇംഗ്ലീഷ് Image Fundamentals Images, regions, piecewise contours, Image Enhancement: contrast ഇംഗ്ലീഷ് അർത്ഥം definition/meaning ഇംഗ്ലീഷ് അർത്ഥം. ഇംഗ്ലീഷ് അർത്ഥം ഇംഗ്ലീഷ്, steps, application ഇംഗ്ലീഷ് അർത്ഥം ഇംഗ്ലീഷ്. Technical terms English-ഇംഗ്ലീഷ് അർത്ഥം ഇംഗ്ലീഷ്.

## enhancement, contrast normalization, Image Smoothing: Temporal Averaging, Mean Filter, Noise

enhancement, contrast normalization, Image Smoothing: Temporal Averaging, Mean Filter, Noise is an official topic in Module 2 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify enhancement, contrast normalization, Image Smoothing: Temporal Averaging, Mean Filter, Noise using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, enhancement, contrast normalization, image smoothing: temporal averaging, mean filter, noise should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                        |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What enhancement, contrast normalization, Image Smoothing: Temporal Averaging, Mean Filter, Noise means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-ഇംഗ്ലീഷ് enhancement, contrast normalization, Image Smoothing: Temporal Averaging, Mean Filter, Noise ഇംഗ്ലീഷ് definition/meaning ഇംഗ്ലീഷ്. ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്, steps, application ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്. Technical terms English-ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്.



## Suppression by Linear Filters, Median and Rank Filters, 2-D Fourier Transform: Continuous

Suppression by Linear Filters, Median and Rank Filters, 2-D Fourier Transform: Continuous is an official topic in Module 2 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Suppression by Linear Filters, Median and Rank Filters, 2-D Fourier Transform: Continuous using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, suppression by linear filters, median and rank filters, 2-d fourier transform: continuous should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Suppression by Linear Filters, Median and Rank Filters, 2-D Fourier Transform: Continuous means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- $\square$  Suppression by Linear Filters, Median and Rank Filters, 2-D Fourier Transform: Continuous  $\square$   $\square$  definition/meaning  $\square$ .  $\square$   $\square$   $\square$ , steps, application  $\square$   $\square$ . Technical terms English- $\square$   $\square$   $\square$ .

## Fourier Transform, Discrete Fourier Transform. Geometric Transformations: Affine

Fourier Transform, Discrete Fourier Transform. Geometric Transformations: Affine is an official topic in Module 2 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Fourier Transform, Discrete Fourier Transform. Geometric Transformations: Affine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, fourier transform, discrete fourier transform. geometric transformations: affine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                            |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Fourier Transform, Discrete Fourier Transform. Geometric Transformations: Affine means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                    |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                          |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-ഫോറിയർ ട്രാൻസ്ഫോം, ഡിസ്ക്രീറ്റ് ഫോറിയർ ട്രാൻസ്ഫോം. Geometric Transformations: Affine ട്രാൻസ്ഫോം ട്രാൻസ്ഫോം definition/meaning ട്രാൻസ്ഫോം. ട്രാൻസ്ഫോം ട്രാൻസ്ഫോം, steps, application ട്രാൻസ്ഫോം ട്രാൻസ്ഫോം. Technical terms English-ഫോറിയർ ട്രാൻസ്ഫോം.

## Transformations, Projective Transformations

Transformations, Projective Transformations is an official topic in Module 2 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Transformations, Projective Transformations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, transformations, projective transformations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                       |
|---------------------|---------------------------------------------------------------------------------------|
| Definition/identity | What Transformations, Projective Transformations means in the official course context |
| Study lens          | Principle → components or stages → result → application                               |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate     |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- $\mathbb{R}$  Transformations, Projective Transformations  
പ്രസ്താവിക്കുന്ന  $\mathbb{R}$  definition/meaning പ്രസ്താവിക്കുന്നു.  $\mathbb{R}$   $\mathbb{R}$   $\mathbb{R}$ ,  
steps, application  $\mathbb{R}$   $\mathbb{R}$   $\mathbb{R}$ . Technical terms English- $\mathbb{R}$   $\mathbb{R}$   
 $\mathbb{R}$ .

## Explain the different image transformation

Explain the different image transformation is an official topic in Module 2 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain the different image transformation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain the different image transformation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                      |
|---------------------|--------------------------------------------------------------------------------------|
| Definition/identity | What Explain the different image transformation means in the official course context |
| Study lens          | Principle → components or stages → result → application                              |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate    |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- Explain the different image transformation techniques. Explain the definition/meaning of image transformation. Explain the steps, application of image transformation. Technical terms English- Explain the image transformation.



## techniques

techniques is an official topic in Module 2 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What techniques means in the official course context                              |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എന്ന techniques പരിചയപ്പെടുത്തുന്ന പാഠ്യം  
definition/meaning പരിചയപ്പെടുത്തുന്നു. പാഠ്യം പാഠ്യം പാഠ്യം, steps, application  
പാഠ്യം പരിചയപ്പെടുത്തുന്നു പാഠ്യം. Technical terms English-എന്ന പാഠ്യം പരിചയപ്പെടുത്തുന്നു.

## Explain the different thresholding algorithms used in

Explain the different thresholding algorithms used in is an official topic in Module 2 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain the different thresholding algorithms used in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain the different thresholding algorithms used in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                 |
|---------------------|-------------------------------------------------------------------------------------------------|
| Definition/identity | What Explain the different thresholding algorithms used in means in the official course context |
| Study lens          | Principle → components or stages → result → application                                         |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate               |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- Explain the different thresholding algorithms used in image processing. Define/meaning of thresholding. Explain the steps, application of thresholding. Technical terms English-Malayalam.

## image processing.

image processing. is an official topic in Module 2 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify image processing. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, image processing. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What image processing. means in the official course context                       |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-ഇംഗ്ലീഷ് image processing. ഇംഗ്ലീഷ് definition/meaning ഇംഗ്ലീഷ്. ഇംഗ്ലീഷ് steps, application ഇംഗ്ലീഷ്. Technical terms English-ഇംഗ്ലീഷ്.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Image Fundamentals Images, regions, piecewise contours, Image Enhancement: contrast enhancement, contrast normalization, Image Smoothing: Temporal Averaging, Mean Filter, Noise Suppression by Linear Filters, Median and Rank Filters, 2-D Fourier Transform: Continuous Fourier Transform, Discrete Fourier Transform. Geometric Transformations: Affine Transformations, Projective Transformations

C03: To expose the students to image feature detection and matching.

Explain the different image transformation

|                                                       |   |
|-------------------------------------------------------|---|
| M3.01                                                 | 8 |
| Understanding                                         |   |
| techniques                                            |   |
| Explain the different thresholding algorithms used in |   |
| M3.02                                                 | 7 |
| Understanding                                         |   |
| image processing.                                     |   |

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Image Transformations: Nearest-Neighbour Interpolation, Bilinear Interpolation, Bicubic

This module is presented from the official syllabus scope for Machine Vision. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Image Transformations: Nearest-Neighbour Interpolation, Bilinear Interpolation, Bicubic
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## Image Transformations: Nearest-Neighbour Interpolation, Bilinear Interpolation, Bicubic

Image Transformations: Nearest-Neighbour Interpolation, Bilinear Interpolation, Bicubic is an official topic in Module 3 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Image Transformations: Nearest-Neighbour Interpolation, Bilinear Interpolation, Bicubic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, image transformations: nearest-neighbour interpolation, bilinear interpolation, bicubic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Image Transformations: Nearest-Neighbour Interpolation, Bilinear Interpolation, Bicubic means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                 |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.



**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-ഇംഗ്ലീഷ് Image Transformations: Nearest-Neighbour Interpolation, Bilinear Interpolation, Bicubic ഇംഗ്ലീഷ് അർത്ഥം definition/meaning ഇംഗ്ലീഷ് അർത്ഥം. ഇംഗ്ലീഷ് അർത്ഥം ഇംഗ്ലീഷ്, steps, application ഇംഗ്ലീഷ് അർത്ഥം ഇംഗ്ലീഷ്. Technical terms English-ഇംഗ്ലീഷ് അർത്ഥം ഇംഗ്ലീഷ്.

## Interpolation, Smoothing to Avoid Aliasing, Projective Image Transformations, Transformations,

Interpolation, Smoothing to Avoid Aliasing, Projective Image Transformations, Transformations, is an official topic in Module 3 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Interpolation, Smoothing to Avoid Aliasing, Projective Image Transformations, Transformations, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, interpolation, smoothing to avoid aliasing, projective image transformations, transformations, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Interpolation, Smoothing to Avoid Aliasing, Projective Image Transformations, Transformations, means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                  |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                        |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-ഇംഗ്ലീഷ് Interpolation, Smoothing to Avoid Aliasing, Projective Image Transformations, Transformations, ഏകീകൃതമായ ഏകീകൃത definition/meaning ഇംഗ്ലീഷ്. ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്, steps, application ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്. Technical terms English-ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്.

## Image Segmentation Thresholding: Global Thresholding, Automatic Threshold Selection,

Image Segmentation Thresholding: Global Thresholding, Automatic Threshold Selection, is an official topic in Module 3 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Image Segmentation Thresholding: Global Thresholding, Automatic Threshold Selection, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, image segmentation thresholding: global thresholding, automatic threshold selection, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Image Segmentation Thresholding: Global Thresholding, Automatic Threshold Selection, means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                        |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                              |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-ഇംഗ്ലീഷ് Image Segmentation Thresholding: Global Thresholding, Automatic Threshold Selection, ഇംഗ്ലീഷ് definition/meaning ഇംഗ്ലീഷ്. ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്, steps, application ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്. Technical terms English-ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്.

## Dynamic Thresholding, Variation Model, morphological operations.

Dynamic Thresholding, Variation Model, morphological operations. is an official topic in Module 3 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Dynamic Thresholding, Variation Model, morphological operations. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, dynamic thresholding, variation model, morphological operations. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Dynamic Thresholding, Variation Model, morphological operations. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                    |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                          |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 3- dynamic Thresholding, Variation Model, morphological operations. definition/meaning . steps, application . Technical terms English- .

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Image Transformations: Nearest-Neighbour Interpolation, Bilinear Interpolation, Bicubic Interpolation, Smoothing to Avoid Aliasing, Projective Image Transformations, Transformations,  
Image Segmentation Thresholding: Global Thresholding, Automatic Threshold Selection,  
Dynamic Thresholding, Variation Model, morphological operations.  
C04:Apply various transform tools like frequency domain and affine transform.  
M4.01 Discuss edge-based approaches to segmentation 7  
M4.02 Explain the technique for texture image segmenting. 7  
Series Test – II 1

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## MODULE 4

# Image Segmentation: Region Growing, Edge Based approaches to segmentation, Graph-Cut,

This module is presented from the official syllabus scope for Machine Vision. Use the topic cards for learning and the source transcript for exact coverage.

As specified

CO-linked

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Image Segmentation: Region Growing, Edge Based approaches to segmentation, Graph-Cut,
- Official duration: As specified
- Official topic entries captured: 13
- The full source excerpt is preserved below for coverage checking.



## Image Segmentation: Region Growing, Edge Based approaches to segmentation, Graph-Cut,

Image Segmentation: Region Growing, Edge Based approaches to segmentation, Graph-Cut, is an official topic in Module 4 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Image Segmentation: Region Growing, Edge Based approaches to segmentation, Graph-Cut, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, image segmentation: region growing, edge based approaches to segmentation, graph-cut, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Image Segmentation: Region Growing, Edge Based approaches to segmentation, Graph-Cut, means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                         |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                               |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-ഇംഗ്ലീഷ് Image Segmentation: Region Growing, Edge Based approaches to segmentation, Graph-Cut, ഏതെങ്കിലും രീതിയിൽ രേഖാപ്പെടുത്തൽ definition/meaning എഴുതേണ്ടതാണ്. ഏതെങ്കിലും രീതിയിൽ രേഖാപ്പെടുത്തൽ, steps, application എഴുതേണ്ടതാണ് രേഖാപ്പെടുത്തൽ രീതിയിൽ. Technical terms English-ൽ എഴുതേണ്ടതാണ് രേഖാപ്പെടുത്തൽ.

## Mean-Shift, MRFs, Texture Segmentation; Object detection.

Mean-Shift, MRFs, Texture Segmentation; Object detection. is an official topic in Module 4 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Mean-Shift, MRFs, Texture Segmentation; Object detection. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, mean-shift, mrfs, texture segmentation; object detection. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------|
| Definition/identity | What Mean-Shift, MRFs, Texture Segmentation; Object detection. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- Mean-Shift, MRFs, Texture Segmentation; Object detection. definition/meaning . steps, application . Technical terms English- .

## Text / Reference

Text / Reference is an official topic in Module 4 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Text / Reference using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, text / reference should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Text / Reference means in the official course context                        |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-  
Text / Reference definition/meaning . . . . .  
definition/meaning . . . . . steps, application  
 . . . . . Technical terms English-  
 . . . . .

## T/R Book Title/Author

T/R Book Title/Author is an official topic in Module 4 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify T/R Book Title/Author using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, t/r book title/author should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What T/R Book Title/Author means in the official course context                   |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-  
T/R Book Title/Author  
definition/meaning .  
 , steps, application  
 . Technical terms English-  
 .



## Carsten Steger, Markus Ulrich, Christian Wiedemann, Machine Vision Algorithms and

Carsten Steger, Markus Ulrich, Christian Wiedemann, Machine Vision Algorithms and is an official topic in Module 4 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Carsten Steger, Markus Ulrich, Christian Wiedemann, Machine Vision Algorithms and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, carsten steger, markus ulrich, christian wiedemann, machine vision algorithms and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Carsten Steger, Markus Ulrich, Christian Wiedemann, Machine Vision Algorithms and means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                     |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                           |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-എന്നു Carsten Steger, Markus Ulrich, Christian Wiedemann, Machine Vision Algorithms and ഏകീകൃതമായ ഏകീകൃത definition/meaning എഴുതേണ്ടതാണ്. എഴുതേണ്ടതാണ് steps, application എഴുതേണ്ടതാണ്. Technical terms English-എന്നു എഴുതേണ്ടതാണ്.

Applications, WILEY-VCH, Weinheim,2008. is an official topic in Module 4 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Applications, WILEY-VCH, Weinheim,2008. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, applications, wiley-vch, weinheim,2008. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Applications, WILEY-VCH, Weinheim,2008. means in the official course context |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-എന്ന Applications, WILEY-VCH, Weinheim,2008.

എന്ന definition/meaning എന്ന. എന്ന എന്ന, steps, application എന്ന എന്ന. Technical terms English-എന്ന എന്ന എന്ന.

## T2 Computer Vision: A Modern Approach, D. A. Forsyth, J. Ponce, Pearson Education, 2003.

T2 Computer Vision: A Modern Approach, D. A. Forsyth, J. Ponce, Pearson Education, 2003. is an official topic in Module 4 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify T2 Computer Vision: A Modern Approach, D. A. Forsyth, J. Ponce, Pearson Education, 2003. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, t2 computer vision: a modern approach, d. a. forsyth, j. ponce, pearson education, 2003. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                    |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What T2 Computer Vision: A Modern Approach, D. A. Forsyth, J. Ponce, Pearson Education, 2003. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                            |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                  |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-എം T2 Computer Vision: A Modern Approach, D. A. Forsyth, J. Ponce, Pearson Education, 2003. ഏകദേശം ഏകദേശം definition/meaning എഴുതണം. എഴുതണം എഴുതണം, steps, application എഴുതണം എഴുതണം എഴുതണം. Technical terms English-എം എഴുതണം എഴുതണം.

Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing, Addition – Wesley is an official topic in Module 4 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing, Addition – Wesley using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing, Addition – Wesley should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing, Addition – Wesley means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                        |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                              |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-റാഫേൽ സി. ഗോണ്ടാലസ് ആൻഡ് റിച്ചാർഡ് ഇ. വുഡ്സ്, ഡിജിറ്റൽ ഇമേജ് പ്രോസസിംഗ്, അഡിഷൻ - വെസലി ഡിഫിനിഷൻ/മീനിംഗ് ഓർഗനൈസേഷൻ. ഓർഗനൈസ്ഡ് ഓർഗനൈസേഷൻ, steps, application ഓർഗനൈസ്ഡ് ഓർഗനൈസേഷൻ. Technical terms English-റാഫേൽ സി. ഗോണ്ടാലസ് ആൻഡ് റിച്ചാർഡ് ഇ. വുഡ്സ്, ഡിജിറ്റൽ ഇമേജ് പ്രോസസിംഗ്, അഡിഷൻ - വെസലി ഡിഫിനിഷൻ/മീനിംഗ് ഓർഗനൈസേഷൻ.



## Publishing Company, New Delhi, 2007.

Publishing Company, New Delhi, 2007. is an official topic in Module 4 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Publishing Company, New Delhi, 2007. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, publishing company, new delhi, 2007. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Publishing Company, New Delhi, 2007. means in the official course context    |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-എസ്സി Publishing Company, New Delhi, 2007.

എസ്സിഎസ്സിഎസ്സി definition/meaning എസ്സിഎസ്സിഎസ്സി. എസ്സിഎസ്സി എസ്സിഎസ്സി,  
steps, application എസ്സിഎസ്സി എസ്സിഎസ്സി. Technical terms English-എസ്സി എസ്സി  
എസ്സിഎസ്സിഎസ്സി.

Shimon Ullman, High-Level Vision: Object recognition and Visual Cognition, A is an official topic in Module 4 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Shimon Ullman, High-Level Vision: Object recognition and Visual Cognition, A using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, shimon ullman, high-level vision: object recognition and visual cognition, a should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Shimon Ullman, High-Level Vision: Object recognition and Visual Cognition, A means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 4-ശിമൺ യുല്മാൻ, High-Level Vision: Object recognition and Visual Cognition, A ശിമൺ യുല്മാൻ definition/meaning ശിമൺ യുല്മാൻ. ശിമൺ യുല്മാൻ ശിമൺ യുല്മാൻ, steps, application ശിമൺ യുല്മാൻ ശിമൺ യുല്മാൻ. Technical terms English-ശിമൺ യുല്മാൻ ശിമൺ യുല്മാൻ.

## Bradford Book, USA, 2000.

Bradford Book, USA, 2000. is an official topic in Module 4 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Bradford Book, USA, 2000. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, bradford book, usa, 2000. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Bradford Book, USA, 2000. means in the official course context               |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-  
definition/meaning  
steps, application  
Technical terms English-  
English

## R.Patrick Goebel, ROS by Example: A Do-It-Yourself Guide to Robot Operating System

R.Patrick Goebel, ROS by Example: A Do-It-Yourself Guide to Robot Operating System is an official topic in Module 4 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify R.Patrick Goebel, ROS by Example: A Do-It-Yourself Guide to Robot Operating System using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, r.patrick goebel, ros by example: a do-it-yourself guide to robot operating system should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What R.Patrick Goebel, ROS by Example: A Do-It-Yourself Guide to Robot Operating System means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                      |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                            |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- R.Patrick Goebel, ROS by Example: A Do-It-Yourself Guide to Robot Operating System definition/meaning . steps, application . Technical terms English-



## **-Volume I, A Pi Robot Production, 2012.**

-Volume I, A Pi Robot Production, 2012. is an official topic in Module 4 of Machine Vision. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify -Volume I, A Pi Robot Production, 2012. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### **Engineering or professional relevance**

In Polytechnic practice, -volume i, a pi robot production, 2012. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### **Topic study guide**

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What -Volume I, A Pi Robot Production, 2012. means in the official course context |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-റൂട്ടിംഗ് -Volume I, A Pi Robot Production, 2012.

രൂട്ടിംഗ് എന്നതിന്റെ definition/meaning എന്തായിരിക്കണം. രൂട്ടിംഗ് എങ്ങനെ പ്രയോജനപ്പെടുന്നു, steps, application എങ്ങനെ രൂട്ടിംഗ് പ്രയോജനപ്പെടുന്നു. Technical terms English-ൽ എങ്ങനെ രൂട്ടിംഗ് പ്രയോജനപ്പെടുന്നു.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.  
Educational explanations below are separate elaboration.

Image Segmentation: Region Growing, Edge Based approaches to segmentation, Graph-Cut, Mean-Shift, MRFs, Texture Segmentation; Object detection.

Text / Reference

| T/R | Book Title/Author                                                                                                           |
|-----|-----------------------------------------------------------------------------------------------------------------------------|
|     | Carsten Steger, Markus Ulrich, Christian Wiedemann, Machine Vision Algorithms and Applications, WILEY-VCH, Weinheim, 2008.  |
| T1  |                                                                                                                             |
| T2  | Computer Vision: A Modern Approach, D. A. Forsyth, J. Ponce, Pearson Education, 2003.                                       |
|     | Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing, Addition – Wesley Publishing Company, New Delhi, 2007.   |
| R1  |                                                                                                                             |
| A   | Shimon Ullman, High-Level Vision: Object recognition and Visual Cognition, Bradford Book, USA, 2000.                        |
| R2  |                                                                                                                             |
|     | R. Patrick Goebel, ROS by Example: A Do-It-Yourself Guide to Robot Operating System -Volume I, A Pi Robot Production, 2012. |
| R3  |                                                                                                                             |

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

# Concept and formula bank

## Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

## Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

## RAPID REVISION

### Diagram and source-transcript library



#### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.



#### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.



#### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.



## Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### OFFICIAL SELF-LEARNING

## Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

### EXAMINATION PREPARATION

## Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

### ACTIVE RECALL

## Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

### Module 1

**Define or explain Image Acquisition, Lenses and Cameras: Pinhole camera, Gaussian Optics, Depth Field,. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Image Acquisition, Lenses and Cameras: Pinhole camera, Gaussian Optics, Depth Field,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Telemetric lenses, Lens aberrations Cameras: CCD cameras, CMOS Camera, Colour cameras:.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Telemetric lenses, Lens aberrations Cameras: CCD cameras, CMOS Camera, Colour cameras:.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Single chip cameras, Three chip cameras, Camera performance parameters: noises, dynamic range,. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Single chip cameras, Three chip cameras, Camera performance parameters: noises, dynamic range,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Camera-Computer interfaces, Digital video signals: camera**

**link, IEEE1394, USB2.0, USB 3. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Camera-Computer interfaces, Digital video signals: camera link, IEEE1394, USB2.0, USB 3*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

## Module 2

**Define or explain Image Fundamentals Images, regions, piecewise contours, Image Enhancement: contrast. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Image Fundamentals Images, regions, piecewise contours, Image Enhancement: contrast*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain enhancement, contrast normalization, Image Smoothing: Temporal Averaging, Mean Filter, Noise. (3 marks)**

**Answer:** Begin with the standard definition or identity of *enhancement, contrast normalization, Image Smoothing: Temporal Averaging, Mean Filter, Noise*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Suppression by Linear Filters, Median and Rank Filters, 2-D Fourier Transform: Continuous. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Suppression by Linear Filters, Median and Rank Filters, 2-D Fourier Transform: Continuous*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Fourier Transform, Discrete Fourier Transform. Geometric Transformations: Affine. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Fourier Transform, Discrete Fourier Transform. Geometric Transformations: Affine*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## Module 3

**Define or explain Image Transformations: Nearest-Neighbour Interpolation, Bilinear Interpolation, Bicubic. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Image Transformations: Nearest-Neighbour Interpolation, Bilinear Interpolation, Bicubic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Interpolation, Smoothing to Avoid Aliasing, Projective Image Transformations, Transformations,. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Interpolation, Smoothing to Avoid Aliasing, Projective Image Transformations, Transformations*,. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Image Segmentation Thresholding: Global Thresholding, Automatic Threshold Selection, (3 marks)**

**Answer:** Begin with the standard definition or identity of *Image Segmentation Thresholding: Global Thresholding, Automatic Threshold Selection*,. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Dynamic Thresholding, Variation Model, morphological operations.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Dynamic Thresholding, Variation Model, morphological operations*.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## Module 4

**Define or explain Image Segmentation: Region Growing, Edge Based approaches to segmentation, Graph-Cut, (3 marks)**



**Answer:** Begin with the standard definition or identity of *Image Segmentation: Region Growing, Edge Based approaches to segmentation, Graph-Cut*,. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Mean-Shift, MRFs, Texture Segmentation; Object detection.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Mean-Shift, MRFs, Texture Segmentation; Object detection*.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Text / Reference. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Text / Reference*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain T/R Book Title/Author. (3 marks)**

**Answer:** Begin with the standard definition or identity of *T/R Book Title/Author*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

## PRACTICE ONLY

# Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

## Module 1

1-mark definition: State the meaning of **Image Acquisition, Lenses and Cameras: Pinhole camera, Gaussian Optics, Depth Field,**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 2

1-mark definition: State the meaning of **Image Fundamentals Images, regions, piecewise contours, Image Enhancement: contrast.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 3

1-mark definition: State the meaning of **Image Transformations: Nearest-Neighbour Interpolation, Bilinear Interpolation, Bicubic.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 4

1-mark definition: State the meaning of **Image Segmentation: Region Growing, Edge Based approaches to segmentation, Graph-Cut,**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### QUICK REVISION

## Exam checklist

### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

### REFERENCE VOCABULARY

## Glossary

Study glossary

| Term              | Meaning in this lesson                                                   |
|-------------------|--------------------------------------------------------------------------|
| Course Outcome    | A capability the student should demonstrate after completing the course. |
| Module            | An official syllabus unit with defined topics and hours.                 |
| CIE               | Continuous Internal Evaluation.                                          |
| ESE               | End Semester Examination.                                                |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

## SOURCE-SUPPORTED REFERENCES

### References and web resources

#### References listed in the official PDF

References are included in the official source PDF.

#### Web resources listed in the official PDF

No separate web-resource heading was detected in the extracted text.

## IN-PAGE SEARCH

### Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 5333B. Verify institutional updates against the current official curriculum.